

Numbering System Conversions

Show Work! Point deduction taken if work not shown

1. Convert 147_{10} to binary

128, 64, 32, 16, 8, 4, 2, 1

(128x1), (64x0), (32x0), (16x1), (8x0), (4x0), (2x1), (1x1)

$$128 + 16 + 2 + 1 = 147^{10}$$

10010011²

2. Convert 63_{10} to binary

128, 64, 32, 16, 8, 4, 2, 1

(128x0), (64x0), (32x1), (16x1), (8x1), (4x1), (2x1), (1x1)

$$32 + 16 + 8 + 4 + 2 + 1 = 63^{10}$$

00111111²

3. Convert 97_{10} to hexadecimal

128, 64, 32, 16, 8, 4, 2, 1

(128x0), (64x1), (32x1), (16x0), (8x0), (4x0), (2x0), (1x1)

$$64 + 32 + 1 = 97^{10}$$

$$97/16 = 6 \text{ R}1$$

$$97^{10} = 61^{16}$$

4. Convert 95_{10} to hexadecimal

128, 64, 32, 16, 8, 4, 2, 1

(128x0), (64x1), (32x0), (16x1), (8x1), (4x1), (2x1), (1x1)

$$64 + 16 + 8 + 4 + 2 + 1 = 95^{10}$$

$$95/16 = 5 \text{ R}15$$

$$95^{10} = 5F^{16}$$

5. Convert $3D_{16}$ to binary

$$3 = 0011, D = 13, 13 = 1101$$

$$(128 \times 0) (64 \times 0) (32 \times 1) (16 \times 1) (8 \times 1) (4 \times 1) (2 \times 0) (1 \times 1)$$

$$3D^{16} = 00111101^2$$

6. Convert 235_{10} to hexadecimal. Then convert hexadecimal value to binary

$$235/16 = 14 \text{ R}11$$

$$14 = E, 11 = B$$

$$235^{10} = EB^{16}. E = 14, 14 = 1110. B = 11, 11 = 1011$$

$$235^{10} = EB^{16} = 11101011^2$$

7. Convert 189_{10} to binary. Then convert binary value to hexadecimal

$$128, 64, 32, 16, 8, 4, 2, 1$$

$$(128 \times 1), (64 \times 0), (32 \times 1), (16 \times 1), (8 \times 1), (4 \times 1), (2 \times 0), (1 \times 1)$$

$$128 + 32 + 16 + 8 + 4 + 1 = 189^{10}$$

$$00111111^2. 1011 = B, 1101 = D$$

$$10111101^2 = BD^{16}$$

$$189^{10} = 10111101^2 = BD^{16}$$

8. Code the following in hexadecimal and then in binary in ASCII. Be sure to review the video on conversions before working on this problem

M=4D O=4F N=4E D=44 A=41 Y=59 (no spaces)

(M, 4 = 0100, D = 1101) (O, 4 = 0100, F = 1111) (N, 4 = 0100, E = 1110) (D, 4 = 0100, 4 = 0100) (A, 4 = 0100, 1 = 0001) (Y, 5 = 0101, 9 = 1001)

Hexadecimal: 4D4F4E444159

Binary ASCII: 010011010100111101001110010001000100000101011001